

SAINTGITS COLLEGE OF ENGINEERING (AUTONOMOUS), KOTTAYAM

First Series Examinations, August 2025
24BSE2113-D Partial Differential Equations, Transforms and Optimization Methods

Duration: 90 minutes

Maximum marks: 50

Part A (Answer all questions)

1. Form a PDE from the relation $z = (x + y)f(x^2 - y^2)$. (5 marks)
2. (a) Form a PDE by elimination of the arbitrary constants a and b from $z = ax^2 + by^2 + ab$. (3 marks)
(b) Solve using Lagrange method, $x(y - z)p + y(z - x)q = z(x - y)$. (7 marks)
3. Using method of separation of variable, solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial y} + u = 0$ where $u(x, 0) = 6e^{-3x}$. (10 marks)

Part B Answer all questions.

4. State all basic assumptions in the solution of one dimensional wave equation of second order and write all possible solutions under these assumptions. (5 marks)
5. A tightly stretched string of length l with fixed ends is initially in equilibrium position. It is set vibrating by giving each point a velocity $g(x) = v_0 \sin^3\left(\frac{\pi x}{l}\right)$. Find the displacement of the string at any time. (10 marks)
6. Derive the one dimensional heat flow through a rectangular rod insulated laterally and kept the two ends at zero temperature with initial heat distribution $f(x)$. (10 marks)